

CS 5588 – Week 11 Hands-On Report

Video LLMs for Real-Time Temporal & Spatial Reasoning

Team-3:

ReliScore: AI Agents for Reliable Medical Text Processing

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For Week-11, we extended our ReliScore medical-intelligence platform by adding a video-analysis module that evaluates patient movement example like during a standard knee-flexion (leg-raise) exercise. This is a common physiotherapy assessment used to measure lower-limb mobility, stability, and neuromuscular control.

Instead of treating the Week-11 work as a separate assignment, we built it directly into our existing multi-agent system, so our main project can now analyze:

- Text
- Images
- 3D anatomy
- Graph-RAG memory
- And now, Video movement patterns

We recorded short clips of seated knee flexion, cropped them for privacy (waist-down only), and used the Week-10 notebook templates as the base for our implementation. We then completed Track A, Track B, and Track C, added our own

clinical-style prompts, and piped all outputs into ReliScore’s knowledge memory system.

2. Track A – Real-Time Loop

We built a small real-time loop inside our ReliScore backend that:

1. Samples windowed batches of frames
2. Sends them to the VLM interface
3. Logs latency, motion intensity, tremor levels, and movement symmetry
4. Streams this information back into ReliScore’s UI module for live feedback

Results (from our Track-A metrics CSV)

Metric	Result
Avg latency	4.48 ms
RTF	909.61
Motion score	6.90
Tremor index	0.276
Symmetry score	0.964

Our interpretation:

The system processed each window **extremely fast** far beyond real time so integrating this into ReliScore doesn’t cause any slowdowns.

The movement itself appears well-controlled: high symmetry, solid motion intensity, and minimal tremor. This confirms that real-time physiotherapy monitoring is feasible inside our platform.

3. Track B – Temporal Reasoning (Reps, Durations, Sequence)

We analyzed the full video clips to extract:

- Total repetitions
- Duration per repetition
- Frequency
- Step-order correctness
- Timing stability

We integrated the resulting structured metrics into the same **GraphRAG memory** used by the text-based portion of ReliScore.

Results (from Track-B temporal CSV)

Metric	Result
Repetition count	4
Avg duration per rep	5.46 sec
Rep frequency	0.178 reps/sec
Latency	904.73 ms
RTF	24.85
MAE (repetition count)	1.0
Step/sequence accuracy	0.80

Our interpretation:

The model correctly identified the general pattern of the exercise. The minor rep-count error matches our manual review one repetition was not completed fully. This track gives ReliScore the ability to generate **session-level summaries**, such as:

“Patient completed 4 controlled repetitions with consistent timing” or
“One repetition showed partial extension.”

This is the first time ReliScore includes **temporal clinical reasoning** rather than just textual analysis.

4. Track C – Spatial Grounding (Joint Localization & Stability)

We processed two higher-resolution knee-flexion videos to extract bounding boxes, track the knee region, and compute:

- Coverage
- IoU change between frames
- Center displacement
- Spatial consistency over time

All outputs were routed into ReliScore’s clinical movement-scoring module.

Results (from our Track-C metrics)

Metric	Result
Avg coverage	0.982
Avg consecutive IoU	0.309
Normalized center displacement	0.121
Latency	4929.83 ms
RTF	4.40

The knee was detected in almost every frame. The IoU variations and center displacement make sense because the knee naturally shifts forward and upward

during the flexion cycle.

These spatial metrics help ReliScore form insights such as:

- “Knee tracking remained stable across the entire exercise.”
- “Joint path shows expected forward arc during extension.”

This gives our platform a visual movement-quality dimension that did not exist before.

5. How we want to integrate Week-11 Work into ReliScore

A. Added a New Video-Assessment Agent

We now have a dedicated “**Video Physiotherapy Agent**” inside ReliScore that:

- Accepts video uploads
- Extracts temporal + spatial movement metrics
- Sends results to the Summary Agent and Reliability Agent
- Stores structured outputs in GraphRAG

B. Reliability Scoring Now Includes Video

Our main project revolves around “reliable medical intelligence.”

Now, reliability is not only text-based—it includes:

- Stability of joint tracking
- Consistency of movement patterns
- Correct sequence execution
- Repetition-timing accuracy

This gives ReliScore a new layer of credibility when analyzing patient progress.

C. Real-Time Dashboard Integration

Track-A metrics are streamed into our existing Streamlit interface, allowing:

- Live rep counting
- Instant motion-symmetry feedback
- On-screen movement quality indicators

This will directly support physiotherapy and tele-rehab use cases.

6. Reflection

This week's work helped us turn ReliScore into a truly multimodal medical system. Before this, our project focused heavily on text, fact-checking, and summarization. Now we have:

- Real-time video understanding
- Clinical temporal reasoning
- Spatial grounding of anatomical regions
- Structured movement-quality scoring

We also learned how to combine these metrics with GraphRAG memory so that future sessions can be compared automatically.

Overall, the task fit well with our vision for ReliScore, and integrating it directly into the platform made the assignment both practical and meaningful for our final project.